SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA0612	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	**Total Marks:**	100 (1 Question × 100 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	JAYAKRISHNA	**Reg. No.:**	192421259
Section / Batch:	1	**Date:**	25/07/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q15	100
		GRAND TOTAL:	**100 Marks**

SECTION A – Technical Presentation

Code of Conduct

I _JAYAKRISHNA.V (192421259) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____JAYAKRISHNA.V___

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

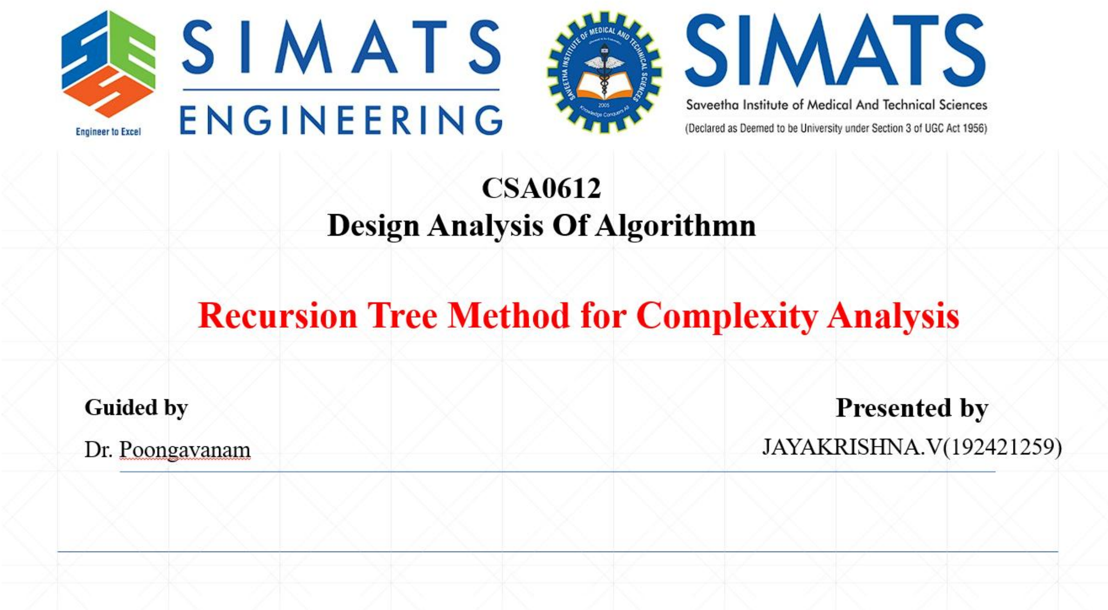
Marks Evaluation:

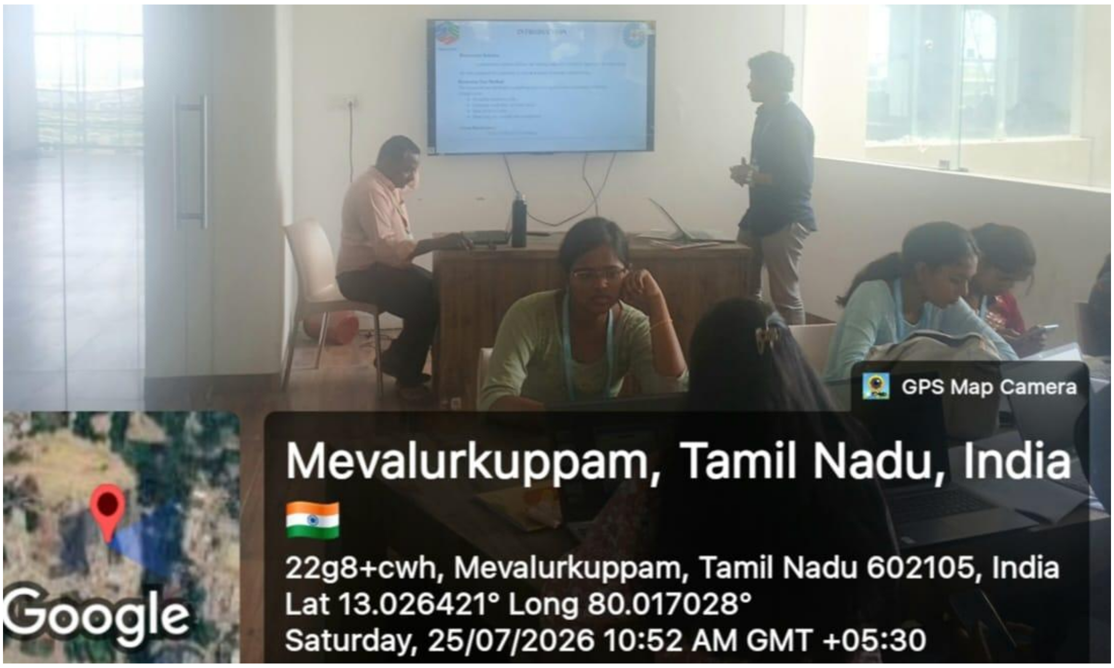
Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q15						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q15. Recursion Tree Method for Complexity Analysis

Analyze the recurrence T(n) = 2T(n/2) + n log n using the recursion tree method. Clearly construct the recursion tree and compute cost at each level. Sum the level costs to derive total complexity. Use diagrams or structured steps for clarity. Interpret the result and compare it with Master Theorem findings.

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